

Divisors and Holomorphic Line Bundles

Preface

Before we begin, two theorems I would like to be kept in mind is the Weierstrass Factorization Theorem and the theorem of Mittag-Leffler.

These two theorems will motivate every single definition that shows up in this document.

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Weil Divisors

Huybrechts 2.3.1. (Analytic hypersurface)

Let X be a complex manifold.

An analytic hypersurface of X is an analytic subvariety $Y \subseteq X$ of codimension one.

A hypersurface is always locally given as the zero set of a single holomorphic function.

Central Question.

How far is a given hypersurface away from being able to be *globally* described by a single holomorphic function?

Or, in terms of algebra:

How far away is an ideal of height n away from being able to be generated by n elements?

For those fluent in commutative algebra, this question is partially answered by Krull's height theorem:

Suppose A is a Noetherian ring and $\mathfrak{p}$ is a minimal prime divisor of $\langle a_1, \dots, a_n \rangle$.
Then $\text{ht}(\mathfrak{p}) \leq n$.

Huybrechts 2.3.2-4. (Weil Divisors)

- A (Weil) divisor D on X is a formal $\mathbb{Z}$ -linear combination of irreducible hypersurfaces that is locally finite.
 - Think about the statement of Mittag-Leffler - the poles of a meromorphic function cannot have any limit points!
They form a free abelian group under the natural addition law, denoted $\text{Div}(X)$.
- Every hypersurface naturally gives a divisor, $Y = \sum [Y_i]$.
 - Notice that all the coefficients are 1 exactly - what would happen if we have a higher degree? Think about the order of a zero of a holomorphic function.
- A divisor is said to be effective if all the coefficients are non-negative.

Huybrechts 2.3.5. (Order w.r.t. a hypersurface)

- Temporarily assume that X is a Riemann surface.

Then the hypersurfaces of X are just closed points $x \in X$.

Recall that the ring $\mathcal{O}_{X,x}$ is a local ring with maximal ideal $\mathfrak{m}_x$.

Then if f is a holomorphic function near x , the order of f with respect to x is defined as the $\mathfrak{m}_x$ -adic valuation of the corresponding germ of f at x ; denoted $\text{ord}_x(f)$ (compare this definition to the order of a zero of a holomorphic function).

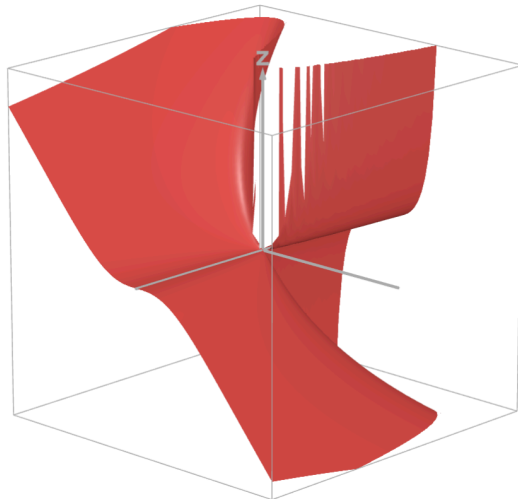
This definition naturally extends to functions that are only meromorphic near x by taking fraction fields.

- We now define the general case for higher dimensional complex manifolds:

If $Y \subseteq X$, then for each $y \in Y$ $\text{ord}_y(f)$ is similarly defined.

Turns out, $\text{ord}_y(f)$ will agree for almost all $y \in Y$ except for potentially a closed set of "irregular" points. We take this to be the definition of $\text{ord}_Y(f)$.

- This is not a very clean definition. In AG, we can define $\text{ord}_Y(f)$ much more straightforwardly by once again taking the $\mathfrak{m}$ -adic valuation of f in the regular local ring $\mathcal{O}_{Y,\eta}$ wrt to the generic point η of Y :
 - For instance, let k be a field and let $X = k^2$, Y be the x -axis and $f = \frac{y^3}{y^2 - x}$.
 - Huybrechts tells us to pick a point $x = (x_0, 0) \in Y$ and compute $\text{ord}_x(f)$. In this case we see that $\text{ord}_x(f) = 3$ for $x_0 \neq 0$, and $\text{ord}_x(f) = 1$ when $x_0 = 0$. So the point $(0, 0)$ is an irregular point, and $\text{ord}_Y(f) = 3$.
 - Hartshorne instead tells us to instead look at the associated schemes $\text{Spec}(k[x, y])$ and $\text{Spec}(k[x, y]/(y))$. We then see that the generic point of Y is $\eta = (0) \in \text{Spec}(k[x, y]/(y))$ (by definition), which corresponds to $(y) \in \text{Spec}(k[x, y])$. Hence $\mathcal{O}_{Y,\eta} = k[x, y]_{(y)}$, which is a DVR with maximal ideal $\mathfrak{m} = (y)k[x, y]_{(y)}$. In this case we see that $\text{val}_{\mathfrak{m}}(f) = \text{val}_{\mathfrak{m}}(y^3) - \text{val}_{\mathfrak{m}}(y^2 - x) = 3$ immediately without having to do any case-by-case analysis.



Huybrechts 2.3.8. (Associated principal divisor)

Suppose that f is meromorphic on X .

Then [Huybrechts 2.3.5. \(Order w.r.t. a hypersurface\)](#) implies that we can naturally associate the divisor:

$$(f) := \sum \text{ord}_Y(f)[Y] \in \text{Div}(X)$$

which can be written as the difference of two effective divisors $(f) = Z(f) - P(f)$, where $Z(f)$ and $P(f)$ are the zero divisor and pole divisor of f respectively.

Any divisor which is given by a function in this manner is known as a principal divisor.

- Think about the statement of Weierstrass factorization for meromorphic functions. Notice that addition of principal divisors correspond to multiplication of the corresponding elements, so taking the divisor of an element can be thought of as taking an "algebraic" logarithm.

Hartshorne p.131 (Divisor class group)

The set of principal divisors form a normal subgroup of $\text{Div}(X)$, denoted $\text{Prin}(X)$.

The quotient group, denoted $\text{Cl}(X)$, is known as the **divisor class group**.

- This is an extremely important invariant of complex manifolds, and schemes in general.
- This group measures the failure of the Mittag-Leffler theorem on complex manifolds.
- It can also be thought of as a measure of the failure of the *flasqueness* of the sheaf of meromorphic functions on X :
 - We say that a sheaf $\mathcal{F}$ is *flasque* if for every pair of open sets $U \subseteq V$, the restriction map $\mathcal{F}(V) \rightarrow \mathcal{F}(U)$ is surjective (for an analogy, think about the Dirichlet problem on a region).
 - Recall from [Huybrechts 2.3.1. \(Analytic hypersurface\)](#) that any analytic hypersurface is locally given by a meromorphic (holomorphic) function. Thus, given $Y \subseteq X$, we get an "open cover" of X by meromorphic functions, which gives us a naturally associated divisor D . Then the residue of D in $\text{Cl}(X)$ can be thought of as a virtual representative of the missing global "meromorphic" function that gives D as its principal associated divisor.
 - This idea of virtual global functions will be explored more in [Cartier Divisors](#).

We will see in [Hartshorne 6.4. \(Class group of projective space\)](#) that a divisor on $\mathbb{P}^1$ is principal has to have degree 0.

- This is true more generally over compact Riemann surfaces:
 - Let f be meromorphic on a compact Riemann surface, and let γ be a loop enclosing all of its poles and zeros. Then we have that $\frac{1}{2\pi i} \oint_{\gamma} \frac{f'}{f} dz = Z(f) - P(f)$.

- But then the integral of f around the interior of γ is the same as its integral around its complement, which has no poles or zeros of f , so $\frac{1}{2\pi i} \oint_{\gamma} \frac{f'}{f} dz = 0$.

Borcherds 35. (Divisors over elliptic curves)

Let $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ be a lattice in $\mathbb{C}$. Then $\mathbb{C}/\Lambda$ naturally inherits the structure of a compact Riemann surface.

- For a divisor to be principal, is it enough for it to have degree 0?
- The answer is no. Turns out, one more condition is needed:

A degree 0 divisor $D = \sum n_i [p_i]$ on $\mathbb{C}/\Lambda$ is principal if and only if $\sum n_i p_i \in \Lambda$.

Proof.

- Necessary:

Let f be meromorphic on $\mathbb{C}/\Lambda$.

We integrate f along the boundary of the fundamental domain of Λ . (We ignore the issue of f having a pole or zero on the boundary - this does not affect the proof structure in any way except for introducing more technicalities.)

Then notice that $\frac{1}{2\pi i} \oint_{\gamma} z \frac{f'}{f} dz = \sum n_i p_i$ exactly.

On the other hand, we see that $z \frac{f'}{f}$ is *almost* doubly periodic, such that the integral along the opposing sides of the fundamental domain of $z \frac{f'}{f}$ cancel out to leave integer multiples of ω_1, ω_2 ($\frac{1}{2\pi i} \int_0^{\omega_1} z \frac{f'(z)}{f(z)} dz - (z + \omega_2) \frac{f'(z+\omega_2)}{f(z+\omega_2)} dz = \omega_2 \cdot \frac{1}{2\pi i} \int_0^{\omega_1} \frac{f'}{f} dz$). In other words, $\sum n_i p_i \in \Lambda$.

- Sufficient:

Suppose we are given $D = \sum n_i [p_i]$ such that $\sum n_i p_i \in \Lambda$.

Recall the Weierstrass σ -function, defined as $\sigma(z) = z \prod_{\lambda \in \Lambda \setminus \{0\}} E_2\left(\frac{z}{\lambda}\right)$, with the property that $\sigma(z + \omega_1) = -e^{2\eta_1(z+\omega_1)} \sigma(z)$ and $\sigma(z + \omega_2) = -e^{2\eta_2(z+\omega_2)} \sigma(z)$.

So one can verify that $f := \prod \sigma(z - p_i)^{n_i}$ is the corresponding meromorphic function with the desired properties. ■

Thus we obtain the SES

$$0 \rightarrow \mathbb{C}/\Lambda \rightarrow \text{Cl}(\mathbb{C}/\Lambda) \xrightarrow{\deg} \mathbb{Z} \rightarrow 0.$$

- As illustrated here, contour integration and the residue theorem play a massive role in the theory of Weil divisors over complex manifolds. This is one of the many reasons for why schemes over the complex numbers enjoy exceptional properties.

Hartshorne 2.6.10. (Divisors over a curve)

Let X be a complete (i.e. proper / closed) nonsingular curve over an algebraically closed field k .

Then the degree of a principal divisor is 0.

Proof.

Let $f \in \mathcal{K}(X)$.

Write $(f) = \sum m_i [P_i] - \sum m_j [Q_j]$ as a difference of its zero divisor and its pole divisor.

WLOG we assume that f is nonconstant, such that $(f) \neq 0$.

Then as k is algebraically closed, f is transcendental over k such that $k[f] \simeq k[x]$.

Now let $U := X \setminus P(f)$ such that $f \in \mathcal{O}_X(U)$, which induces an inclusion of rings $k[f] \hookrightarrow \mathcal{O}_X(U)$.

This gives us a morphism $U = \text{Spec}(\mathcal{O}_X(U)) \rightarrow \text{Spec}(k[f]) \simeq k^1$.

We can do the same construction with $\frac{1}{f}$ instead to get another morphism

$X \setminus Z(f) \rightarrow \text{Spec}(k[\frac{1}{f}])$.

We can glue these two maps together naturally to get a morphism $\varphi : X \rightarrow \mathbb{P}^1$.

But then $\varphi^*([0]) = \sum m_i [P_i]$ and $\varphi^*([\infty]) = \sum m_j [Q_j]$ so we get that

$\deg(f) = \deg \varphi^*([0] - [\infty]) = (\deg \varphi) \cdot (\deg([0] - [\infty])) = 0$ (on this step we invoke Hartshorne 2.6.9 without proof to commute $\deg$ with φ^*) we are done. ■

An Interlude In Number Theory

It turns out that the **divisor class group** has incredible arithmetic applications.

Definition. (Number field)

A number field is a finite degree extension of $\mathbb{Q}$.

If K is a number field, then the subring of all algebraic integers (elements of K that are integral over $\mathbb{Z}$) in K is known as the ring of integers of K , denoted $\mathcal{O}_K$. Equivalently, it is the integral closure of $\mathbb{Z}$ in K .

- The relationship between $\mathcal{O}_K$ and K is very much like that between $\mathbb{Z}$ and $\mathbb{Q}$.
 - When $K = \mathbb{Q}(\zeta_n)$, $\mathcal{O}_K = \mathbb{Z}[\zeta_n]$. There is no easy proof for this fact, and involves a *lot* of algebra.
 - When $K = \mathbb{Q}(\sqrt{d})$ for $d \in \mathbb{Z} \setminus \mathbb{Z}^2$, $\mathcal{O}_K = \mathbb{Z}[\sqrt{d}]$ if $d \equiv 2, 3 \pmod{4}$ and $\mathcal{O}_K = \mathbb{Z}[\frac{1+\sqrt{d}}{2}]$ if $d \equiv 1 \pmod{4}$. This is not too difficult to show.
 - By the Primitive element theorem, any number field has the form $K = \mathbb{Q}(\alpha)$ for some α . However, $\mathcal{O}_K$ is generally *not* of the form $\mathbb{Z}[\alpha]$.
- As Gauss's lemma suggests, it turns out that Diophantine equations in K can be solved by solving a corresponding equation in $\mathcal{O}_K$.
- However, the solvability of polynomials of $\mathcal{O}_K$ is **entirely dependent** on $\mathcal{O}_K$ being a UFD. Problem is, $\mathcal{O}_K$ is in general not a UFD.

- Kummer famously tried fixing this by introducing *ideal numbers*, which is where the word "ideal" comes from.
- That gives us a very strong hint as to how this problem will be fixed.

Theorem. (Fractional ideals)

Suppose that D is an integral domain with $\{0\} \neq \mathfrak{a} \triangleleft D$ a finitely generated ideal and $F := Q(D)$. Then TFAE:

1. $\mathfrak{a}$ is projective.
2. There exists $c_1, \dots, c_n \in F, a_i \in \mathfrak{a}$ such that $c_i \mathfrak{a} \subseteq D$ and $\sum_{i=1}^n c_i a_i = 1$.
3. There exists a finitely generated fractional ideal M such that $\mathfrak{a}M = D$.
4. $\mathfrak{a}$ is invertible in $\text{Cl}(D)$; where

$\text{Cl}(D) := \{0 \neq M \subseteq F \mid M \text{ is a finitely generated } D\text{-submodule}\}$, which forms a monoid with identity D , operation given by multiplication and subgroup $\text{Princ}(D)$.

Proof.

Write $\mathfrak{a} = \langle a_1, \dots, a_n \rangle$. Then there exists a surjective $\phi : D^n \rightarrow \mathfrak{a}$ given by $e_i \mapsto a_i$, such that $\mathfrak{a}$ is projective iff the induced SES splits. Hence let $\psi : \mathfrak{a} \rightarrow D^n$ be such that $\phi \circ \psi = \text{id}_{\mathfrak{a}}$.

Now let $S := D \setminus \{0\}$ such that $S^{-1}D = F$.

Then $S^{-1}\psi : S^{-1}\mathfrak{a} \rightarrow (S^{-1}D)^n$ is an F -linear map where $S^{-1}\mathfrak{a} = F$ and $(S^{-1}D)^n = F^n$.

Then $S^{-1}\psi$ must send $a \in \mathfrak{a}$ to $a\vec{c}$ for some $\vec{c} \in F^n$.

In particular, the SES splits iff there exists $c_i \in F$ such that $a = \phi(\psi(a)) = a \sum_{i=1}^n c_i a_i$ for all $a \in \mathfrak{a}$; such that $\sum_{i=1}^n c_i a_i = 1$ as D is an integral domain. Moreover $c_i \mathfrak{a} \subseteq D$ as $c_i \in \psi(\mathfrak{a}) \subseteq D$.

Moreover, M can be taken to be the D -submodule of F generated by the c_i 's, such that it is a fractional ideal with $\mathfrak{a}M = D$ as $1 \in \mathfrak{a}M$. ■

Definition. (Dedekind domains)

A Dedekind domain is an integral domain R such that:

1. R is Noetherian;
2. R is integrally closed;
3. $\dim(R) = 1$, i.e. all nonzero prime ideals are maximal.

It turns out that every nonzero ideal in a Dedekind domain can be uniquely written as a finite product of prime ideals.

- For instance, $\mathbb{Z}[\sqrt{-5}] = \mathcal{O}_{\mathbb{Q}(\sqrt{-5})}$ is not a UFD because $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$. When we pass into ideals, this is fixed via

$$\begin{aligned}
(6) &= \underbrace{(2, 1 + \sqrt{-5})(2, 1 - \sqrt{-5})}_{(2)} \cdot \underbrace{(3, 1 + \sqrt{-5})(3, 1 - \sqrt{-5})}_{(3)} \\
&= \underbrace{(2, 1 + \sqrt{-5})(3, 1 + \sqrt{-5})}_{(1+\sqrt{-5})} \cdot \underbrace{(2, 1 - \sqrt{-5})(3, 1 - \sqrt{-5})}_{(1-\sqrt{-5})}
\end{aligned}$$

Proof.

We first prove the following weaker statement:

- For all $0 \neq \mathfrak{a} \subseteq R$, there exists $\mathfrak{p}_i \in \text{Spec}(R)$ such that $\mathfrak{p}_1 \cdots \mathfrak{p}_r \subseteq \mathfrak{a}$.

Suppose to the contrary that there exists an ideal $\mathfrak{a}$ that cannot be written as a finite product of prime ideals.

As R is Noetherian, WLOG assume that $\mathfrak{a}$ is maximal wrt this property.

Then $\mathfrak{a}$ necessarily cannot be prime, such that there exists $x, y \in R \setminus \mathfrak{a}$ such that $xy \in \mathfrak{a}$.

Then $\mathfrak{a} \subsetneq (\mathfrak{a}, x)$ and $\mathfrak{a} \subsetneq (\mathfrak{a}, y)$ such that $\mathfrak{p}_1 \cdots \mathfrak{p}_s \subseteq (\mathfrak{a}, x)$ and $\mathfrak{q}_1 \cdots \mathfrak{q}_t \subseteq (\mathfrak{a}, y)$.

Then $\mathfrak{p}_1 \cdots \mathfrak{p}_s \mathfrak{q}_1 \cdots \mathfrak{q}_t \subseteq (\mathfrak{a}, x) \cap (\mathfrak{a}, y) \subseteq \mathfrak{a}$.

Smart applications of [Theorem. \(Fractional ideals\)](#) implies the statement. ■

Definition. (Ideal class group)

Let R be a Dedekind domain.

Then the set of ideals of R is isomorphic to the monoid $\bigoplus_{\mathfrak{p} \in \text{Spec}(R)} \mathbb{N}$.

Moreover, when we include fractional ideals, we get a group that is isomorphic to

$$\bigoplus_{\mathfrak{p} \in \text{Spec}(R)} \mathbb{Z}.$$

Very suggestively, we denote this group by $\text{Cl}(R)$, or $\text{Cl}(K)$ in the case of a number field.

Let $\text{Prin}(R) := \{(x) \mid x \in K^\times\} \simeq K^\times / R^\times$ be the set of principal ideals *and* principal fractional ideals.

Then the **ideal class group** of R is $\text{Cl}(R) := \text{Div}(R) / \text{Prin}(R)$.

The **ideal class group** of R is exactly the **divisor class group** on the affine scheme $\text{Spec}(R)$, and is a measure of the *failure of unique factorization* of R .

- In particular, we have that $h_K = 1 \iff \mathcal{O}_K$ is a UFD $\iff \mathcal{O}_K$ is a PID.
- Also notice that the "proper" ideals of R (known as integral ideals) are the arithmetic equivalent of effective divisors.
- Over $\text{Spec}(\mathbb{Z})$, for instance, $\frac{7}{18}$ can be thought of as the "meromorphic function" that has associated principal divisor $[7] - [2] - 2[3]$.

Even more than that, it miraculously turns out that $\text{Cl}(K)$ is a finite group for all number fields K .

The size of this group is known as the **class number** of K , denoted h_K , and is a very important invariant of number fields.

- By Lagrange's theorem, if $I \triangleleft \mathcal{O}_K$, then I^{h_K} is always principal.
 - It turns out that $h_{\mathbb{Q}(\sqrt{-5})} = 2$, and so for instance $(2, 1 + \sqrt{-5})^2 = (2)$.
-

Two remarks.

1. If anyone is familiar of the construction of the one point compactification of a topological space using ultrafilters, the same philosophy is going on here.
 - In some sense, the deficiency of $\mathcal{O}_K$ is not a true deficiency. $\mathcal{O}_K$ "knows" what elements it is missing to fix the problem, but it doesn't have enough elements to be able to "fill in the gaps".
 - So we replace the elements of $\mathcal{O}_K$ with special elements of $2^{\mathcal{O}_K}$ (ideals), which gives $\mathcal{O}_K$ the freedom to maneuver.
2. The name "module" was introduced by Dedekind in his work on ideals and number fields.
 - The following paragraph an excerpt from [this](#) math stack exchange post:

Dedekind notes that the "collection of all multiples of α " satisfies the conditions of being nonempty, closed under differences, and absorption of multiplication, and so this allows you to use "modular arguments" (as in $a \equiv b \pmod{\alpha}$). So he called them modules, because you could "mod out" by them and do modular arithmetic.

Cartier Divisors

Two problems.

When working with Weil divisors we run into two problems:

1. Analytic hypersurfaces do not generalize well to arbitrary schemes:
 - For the degree of a function wrt to a hypersurface to be well-defined, we need that the hypersurface be well-behaved in certain ways.
 - This is not an issue for complex manifolds, but it already becomes a problem even over irreducible complex varieties, due to potential singular points (see [Hartshorne 6.5.2. \(Affine quadratic cone\)](#)).
2. We really care about the associated function more than the divisor:
 - When we have a divisor, an immediate question we almost always ask is, "is this principal?". If the answer is yes, we swiftly replace the divisor with its associated function.
 - Is it possible to "cut out the middleman"?
 - Also recall what was said previously in [Hartshorne p.131 \(Divisor class group\)](#).
"The residue of nonprincipal divisors modulo $\text{Prin}(X)$ can be thought of virtual

representatives of certain missing global "meromorphic" functions."

- Is it possible to define these "missing" global functions more concretely?

For inspiration, let us think about Riemann surfaces:

- Riemann surfaces are locally homeomorphic to $\mathbb{C}$; yet we actually do not want these local homeomorphisms to pass into a global homeomorphism with $\mathbb{C}$.
 - Indeed, the only Riemann surface that is globally homeomorphic to $\mathbb{C}$ is just $\mathbb{C}$ - a triviality.
- We prevent this from happening via the clutching maps: they make "adjacent" pieces of $\mathbb{C}$ glue together coherently without allowing it to happen "too much".

Recall from the very start that a hypersurface is locally given by holomorphic equations.

- Hence, on different patches U_α and U_β you get holomorphic functions f_α and f_β . On $U_\alpha \cap U_\beta$, they necessarily define the same zero set (namely Y), but are not necessarily the same functions.
 - However, the Weierstrass Factorization Theorem comes to the rescue - we know that two entire functions with the same zeros up to multiplicity must differ exactly by a nowhere-zero holomorphic factor, i.e. a unit in the ring of holomorphic functions.
 - What if we could treat these units as our clutching maps?
This motivates our second definition of a divisor.

Hartshorne p.141 (Cartier Divisors)

Let X be a complex manifold.

Recall that $\mathcal{O}_X$ is the ring of holomorphic functions on X , and $\mathcal{K}_X$ is the ring of meromorphic functions on X .

A Cartier divisor on X is an element of $\mathcal{K}_X^*/\mathcal{O}_X^*$.

- More generally, both $\mathcal{O}_X^*$ and $\mathcal{K}_X^*$ naturally induce a sheaf on X . Then a Cartier divisor is equivalently a global section of the quotient sheaf $\mathcal{K}_X^*/\mathcal{O}_X^*$.

In more concrete terms, a Cartier divisor consists of the following data:

- An open cover U_α of X ;
- A collection of local meromorphic functions f_α , restricted to U_α ; such that
- f_α and f_β differ by a local unit on $U_\alpha \cap U_\beta$.

The group of Cartier divisors is denoted $\text{CaCl}(X)$.

A Cartier divisor is said to be effective if all the f_α 's are holomorphic.

- Verify that this corresponds to the definition of effective Weil divisors using the proof below!

On a scheme satisfying certain nice conditions, the notions of Cartier and Weil divisors coincide.

In particular, on complex manifolds, they define the same object.

- This is captured by the following theorem of Huybrechts:

Huybrechts 2.3.9. (Equivalence of Weil and Cartier divisors)

There exists a natural isomorphism

$$H^0(X, \mathcal{K}_X^* / \mathcal{O}_X^*) \cong \text{Div}(X).$$

Proof.

Recall that $H^0(X, \mathcal{K}_X^* / \mathcal{O}_X^*) := \Gamma(X, \mathcal{K}_X^* / \mathcal{O}_X^*)$ is just the ring of global sections.

So an element $H^0(X, \mathcal{K}_X^* / \mathcal{O}_X^*)$ is exactly a cover (f_α, U_α) as described in [Hartshorne p.141 \(Cartier Divisors\)](#). We forgo the cohomological language for the rest of the proof.

If $Y \subseteq X$ is an irreducible hypersurface such that $Y \cap U_\alpha \cap U_\beta \neq \emptyset$ then $\text{ord}_Y(f_\alpha) = \text{ord}_Y(f_\beta)$ necessarily as they differ by a unit. Hence $\text{ord}_Y(f)$ is well-defined, and $(f) = \sum \text{ord}_Y(f)[Y] \in \text{Div}(X)$. It is easy to verify that this is a group homomorphism.

We show that it is bijective via construction of the inverse.

Let $D = \sum a_i[Y_i] \in \text{Div}(X)$.

Then by definition there exists an open cover U_α of X such that D is finite on each U_α .

By restricting the open sets further if necessary, we can replace each divisor $[Y_i]$ on U_α with its corresponding local root locus $g_{i\alpha}$. Hence we can define $f_\alpha := \prod_i g_{i\alpha}^{a_i}$.

Since $g_{i\alpha}$ and $g_{i\beta}$ on $U_\alpha \cap U_\beta$ define the same irreducible hypersurface, they only differ by an element in $\mathcal{O}^*(U_j \cap U_k)$. So the functions f_j glue to an element $f \in H^0(X, \mathcal{K}_X^* / \mathcal{O}_X^*)$.

One can verify that this is an inverse. ■

- This proof is quite abstract, but it has a very concrete geometric interpretation.

Take $X = \mathbb{C}$.

So a divisor $\sum a_i[p_i]$ is just a collection of points p_i in $\mathbb{C}$ with no limit points, and we want to find a "virtual" meromorphic function with poles at p_i of order a_i exactly (assume for simplicity that $a_i < 0$).

- The Mittag-Leffler theorem immediately tells us there exists a meromorphic function f with exactly that desired property, but for various reasons this is not actually ideal. For algebraic applications, we would very much prefer if the function is a rational, and not meromorphic. So let us try and apply this philosophy to modify f into a form that is more amenable to algebra in this manner.

Take an open cover U_α of $\mathbb{C}$ such that each U_α contains finitely many points p_i , say $\{p_{j\alpha}\}$.

Then, on U_α , we get that $f = e^g \prod_j (z - p_{j\alpha})^{a_{j\alpha}}$, where $\prod_j (z - p_{j\alpha})^{a_{j\alpha}}$ is a *finite* product; i.e. a rational function. So the e^g 's in this case behave like the clutching maps that help transition the behaviour of f between different rational functions.

- The Mittag-Leffler theorem can now be phrased algebraically:

$$H^0(\mathbb{C}, \mathcal{K}_{\mathbb{C}}^* / \mathcal{O}_{\mathbb{C}}^*) = 0.$$

We would like to define the pullback of morphisms on the group of Weil divisors.

The main issue is that, in general, the pullback of an analytic hypersurface is not an analytic hypersurface. There are two ways to resolve this issue:

1. Assume some nice conditions on the morphism, such that we can decompose the pullback of a hypersurface into a union of hypersurfaces.
2. Work with Cartier divisors instead, as the pullback of sections is automatically well-defined without further work. (This is illustration of a situation where Cartier divisors are favoured over Weil divisors.)

So we get the following theorem:

Huybrechts 2.3.12. (Pullback on divisors)

If $f : X \rightarrow Y$ is a holomorphic morphism of connected manifolds that is *dominant*, i.e. $f(X)$ is dense in Y , then there is a naturally induced pullback $f^* : \text{Div}(Y) \rightarrow \text{Div}(X)$.

Recall from [Definition. \(Ideal class group\)](#) that if K is a number field, then $\mathcal{O}_K$ is a UFD if and only if $\text{Cl}(K)$ is trivial.

We see that a similar result holds in general:

Hartshorne 2.6.2.

Suppose that A is a Noetherian integrally closed ring.

Then $\text{Cl}(\text{Spec}(A))$ is trivial if and only if A is a UFD.

Proof.

We assume the following two lemmas:

1. A Noetherian integral domain is a UFD if and only if every prime ideal of height 1 is principal.
 - See Matsumura [2, p.141] or Bourbaki [1, Ch. 7.3].
 - This will also follow as a corollary of theorems you will see later in 200C.

2. Suppose that A is an integral domain with $F := Q(A)$.

Then the integral closure of A in F is equal to $\bigcap_{\text{ht } \mathfrak{p}=1} A_{\mathfrak{p}}$.

- This theorem will also be proven in 200C.

- $(\Leftarrow)$: Suppose that A is a UFD.

A irreducible codimension 1 subvariety of $\text{Spec}(A)$ is just a height one prime ideal.

So by lemma (1) we can write $\mathfrak{p} = (f)$ as an ideal. Then $[\mathfrak{p}] = (f)$ is a principal divisor. So every divisor is principal.

- $(\Rightarrow)$: Suppose that $\text{Cl}(\text{Spec}(A)) = 0$.

Let $\mathfrak{p}$ be a height 1 prime ideal, such that $[\mathfrak{p}]$ is given by $f \in Q(A)$ as a principal divisor.

Then by definition $\text{val}_{\mathfrak{p}}(f) = 1$ and $\text{val}_{\mathfrak{q}}(f) = 0$ for all other height 1 prime ideals $\mathfrak{q}$.

Notice that lemma (2) implies that all effective divisors are elements of A .

In particular $f \in A \cap \mathfrak{p}A_{\mathfrak{p}} = \mathfrak{p}$.

If $g \in \mathfrak{p}$, then we have that $(g) = (f) + D$ for some effective divisor $D = (h)$, $h \in A$.

Then $g = hf$, i.e. $g \in Af$. So f generates $\mathfrak{p}$, and by lemma (1) we are done.

Hartshorne 2.6.4. (Class group of projective space)

Consider $\mathbb{P}^n$ as a projective variety.

For any divisor $D = \sum n_i[Y_i]$, let $\deg D := \sum n_i \deg Y_i$, where $\deg Y_i$ is the degree of the hypersurface Y_i (note that this is a well-defined element of $\mathbb{Z}$ as the compactness of projective space forces every divisor to be finite). Let H be the hyperplane $x_0 = 0$. Then

1. If $\deg D = d$, then $D \sim d[H]$.
2. For any $f \in \mathcal{K}^*$, $\deg f = 0$.
3. The induced map $\deg : \text{Cl}(X) \rightarrow \mathbb{Z}$ is an isomorphism.

Proof.

1. Write $D = Z(D) - P(D)$ as a difference of effective divisors. Then we can write $Z(D) = (f)$ and $P(D) = (g)$ as irreducible hypersurfaces correspond to homogeneous prime ideals of height 1 (see [Hartshorne 6.2.](#)).
Hence $D - d[H] = (\frac{f}{x_0^d g})$ is a principal divisor.
2. Any rational function on $\mathbb{P}^n$ is the quotient of two homogeneous polynomials of the same degree.
3. Follows from (1) and (2), and by considering the fact that $\deg H = 1$. ■

Hartshorne 2.6.5.

Let X be a Noetherian integral separated scheme which is regular in codimension one.

Let Z be a proper closed subset of X , and let $U = X \setminus Z$. Then

1. There exists a surjective homomorphism $\text{Cl}(X) \rightarrow \text{Cl}(U)$ given by $[Y] \mapsto [Y \cap U]$, ignoring the Y 's for which $Y \cap U$ is empty;

2. If the codimension of Z in X is at least 2 then the above homomorphism is an isomorphism;
3. If Z is an irreducible subset of codimension 1, then there exists a SES

$$\mathbb{Z} \rightarrow \text{Cl}(X) \rightarrow \text{Cl}(U) \rightarrow 0$$

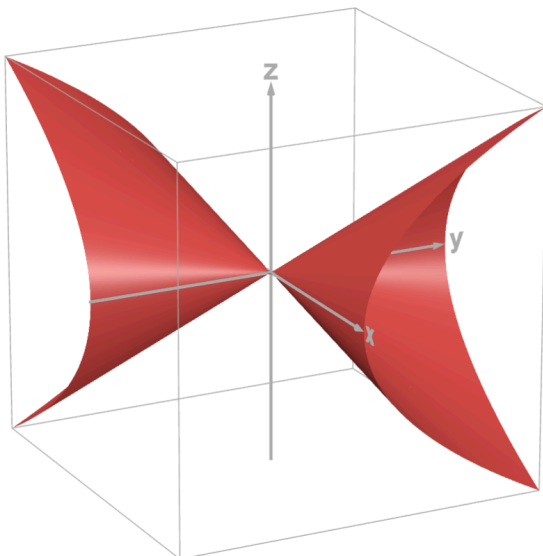
where $\mathbb{Z} \rightarrow \text{Cl}(X)$ is given by $1 \mapsto [Z]$.

Proof.

1. U is an open set of X so $Y \cap U$ is open in Y . Open subsets of irreducible sets are still irreducible. It also necessarily has codimension 1 in U , so $[Y \cap U]$ is a well-defined divisor.
That the map is well-defined from $\text{Div}(X) \rightarrow \text{Div}(U)$ is clear as both are free abelian groups. It suffices to check that it factors through the principal divisors. But then that is also immediate.
 2. The groups $\text{Div}(X)$ and $\text{Cl}(X)$ depend only on subsets of codimension 1, so removing a closed subset of codimension at least 2 does not affect anything.
 3. The kernel of $\text{Cl}(X)$ are all the divisors whose support is contained entirely in Z . If Z is itself irreducible of codimension 1, this can only be the divisor $[Z]$ and its multiples. ■
- Combining this result with [Hartshorne 6.4. \(Class group of projective space\)](#), we get this result (Hartshorne 6.5.1.):
 - Let Y be an irreducible curve of degree d in $\mathbb{P}^2$. Then $\text{Cl}(\mathbb{P}^2 \setminus Y) = \mathbb{Z}/d\mathbb{Z}$.

Hartshorne 6.5.2. (Affine quadratic cone)

Consider the complex variety X given by $\mathbb{C}[x, y, z]/(xy - z^2)$.
This is an affine quadratic cone in $\mathbb{C}^3$.



We show that $\text{Cl}(X) = \mathbb{Z}/2\mathbb{Z}$.

Proof.

Let Y be given by $y = z = 0$.

Y is given set-theoretically on X by the function $y = 0$, where the divisor of y is $(y) = 2[Y]$.

So $X - Y$ is given by $\text{Spec}(k[x, y, z]/(xy - z^2)_{(y)}) \simeq \text{Spec}(k[y, y^{-1}, z])$.

But then $k[y, y^{-1}, z]$ is a UFD, so its class group is trivial by [Hartshorne 6.2.](#)

Hence by the SES in [Hartshorne 6.5.](#) we get that $\text{Cl}(X) \hookrightarrow \mathbb{Z}/2\mathbb{Z}$, generated by $[Y]$.

It suffices to check that $[Y]$ is not a principal divisor. But then that follows as (y, z) is not principal in $k[x, y, z]/(xy - z^2)$. ■

We can make two very interesting observations from this example.

1. The fact that there appears a factor of 2 in $(y) = 2[Y]$ is reflecting the fact that the variety has a singular point at the origin such that the algebraic multiplicity is higher than the apparent geometric multiplicity.
 - Nontrivial class groups detect the existence of singular points on the curve.
2. Y is an example of a Weil divisor that does not have a corresponding Cartier divisor. This is because any rational function on X with a zero on Y will necessarily pick up a factor of 2 along $[Y]$ exactly due to the existence of the singular point.
 - Singular points introduce "extra" Weil divisors that has low multiplicity that cannot be reached by Cartier divisors.
 - This gives good motivation for why the resolution of singular points is a problem we care about. Nathan will be covering how to do this on this talk on blowups.

When talking about holomorphic and meromorphic functions on manifolds, it is tempting to treat them as actual holomorphic and meromorphic functions on $\mathbb{C}$. However, one must remember that such functions are only locally well-defined on open sets - when working with them, keep in mind of the gluing conditions and make sure they remain compatible.

We give a theorem illustrating this - also see [Vakil Ch 14.1. \(Möbius bundle\)](#).

Huybrechts 2.3.31.

Let $f \in K(X)$ be a meromorphic function on a curve X .

Then the induced map $X \setminus P(f) \rightarrow \mathbb{C}$ extends naturally to a holomorphic map $X \rightarrow \mathbb{P}^1$.

Proof.

Let U_α be an open cover of X such that $f = \frac{g_\alpha}{h_\alpha}$ on U_α , where g_α and h_α are holomorphic functions that share no common zero.

Hence the map $U_\alpha \rightarrow \mathbb{P}^1$ given by $x \mapsto (g_\alpha(x) : h_\alpha(x))$ is well-defined.

This glues to a well-defined holomorphic map $X \setminus P(f) \rightarrow \mathbb{P}^1$.

To see that it extends holomorphically to $X \rightarrow \mathbb{P}^1$, it suffices to notice that any rational function on $\mathbb{C}$ admits a holomorphic continuation to $\mathbb{C}_\infty \simeq \mathbb{P}^1$. ■

We say that two divisors are linearly equivalent if they are the same modulo $\text{Prin}(X)$.

We give an example to explore this terminology:

Hartshorne 6.10.2. (Group law on elliptic curves)

Let X be the nonsingular cubic curve $y^2z = x^3 - xz^2$ in $\mathbb{P}^2$.

Note that this is a dimension 2 variety, so the irreducible codimension 1 subvarieties are just points.

Let $\text{Cl}^\circ(X)$ denote the kernel of the degree map $\text{Cl}(X) \rightarrow \mathbb{Z}$.

We show that this group is nontrivial, and in fact in correspondence with the set of (closed) points on the curve; which gives X the structure of an (abelian) group variety.

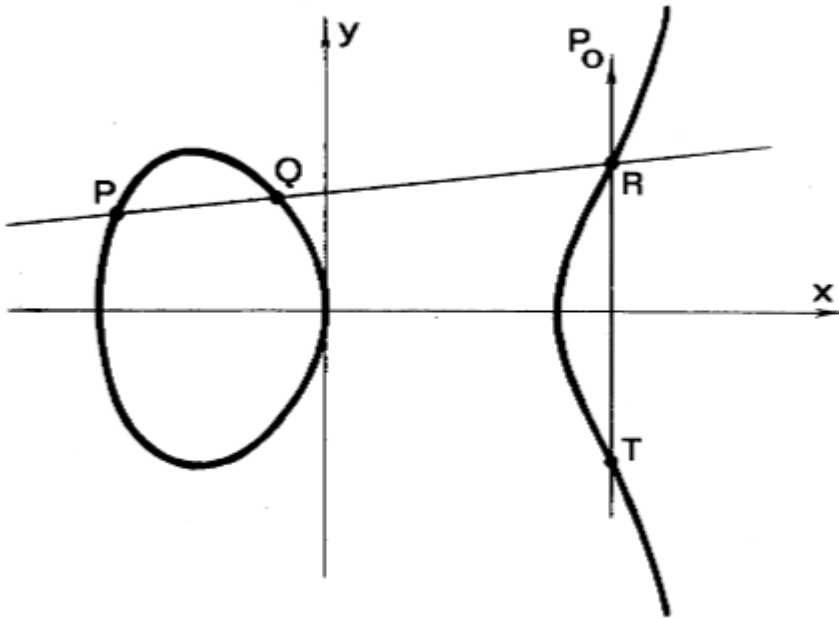


Figure 9. The group law on a cubic curve.

Let $P_0 = (0, 1, 0)$ be the "point at infinity". Its tangent line is $z = 0$, which corresponds to the divisor $3[P_0]$. Any line L in $\mathbb{P}^2$ will meet X at three potentially nondistinct points P, Q, R . Note that by [Hartshorne 6.4. \(Class group of projective space\)](#) we get that $P + Q + R \sim 3[P_0]$ in $\text{Cl}(X)$.

Now let $P \in X$ be a closed point. To this P we associate the divisor $[P] - [P_0]$.

This map has to be injective; as otherwise if there exists $P \neq Q$ such that $[P] - [Q] = (f)$ is principal, by [Huybrechts 2.3.31.](#) we get a naturally induced map $f : X \rightarrow \mathbb{P}^1$, locally given by rational functions, such that it must be surjective. But then this is not possible (e.g. incompatible fundamental groups).

We show the map surjects onto $\text{Cl}^\circ(X)$. We do this by reducing to the case of an effective divisor, then show inductively that we can reduce the degree to 1.

1. Reduction to an effective divisor:

Let $D = \sum n_i [P_i] \in \text{Cl}^\circ(X)$ such that $\sum n_i = 0$. Then $D = \sum n_i ([P_i] - [P_0])$.

If R is any point on X , by the reasoning above we get that $[R] - [P_0] \sim -([T] - [P_0])$. Hence by replacing the terms with negative coefficients, we can WLOG assume that $n_i > 0$ and D is an effective divisor.

2. Induction on degree:

Once again we have that $[P] + [Q] + [R] \sim [P_0] + [R] + [T] \sim 3[P_0]$.

So $([P] - [P_0]) + ([Q] - [P_0]) \sim [T] - [P_0]$. ■

One can further show that the addition law determines a morphism of schemes $X \times X \rightarrow X$, and the inverse law determines a morphism $X \rightarrow X$. Thus X is endowed with the structure of a group variety.

Invertible Sheaves

Hartshorne p.109 (Invertible Sheaves)

An $\mathcal{O}_X$ -module $\mathcal{M}$ is free if it is isomorphic to a direct sum of copies of $\mathcal{O}_X$.

It is locally free if X can be covered by open sets U for which $\mathcal{M}|_U$ is isomorphic to a free $\mathcal{O}_X|_U$ -module. Said isomorphisms are called *trivializations* of $\mathcal{M}$ on U .

In which case the rank of $\mathcal{M}$ on U is naturally defined.

- If X is connected, the rank is a locally free sheaf the same everywhere.

A locally free sheaf of rank 1 is called an invertible sheaf.

Hartshorne 2.6.12. (Picard group)

If $\mathcal{L}$ and $\mathcal{M}$ are invertible sheaves, then $\mathcal{L} \otimes \mathcal{M}$ is also an invertible sheaf.

- Recall that $\mathcal{L} \otimes \mathcal{M}$ is (the sheafification of the presheaf) given by

$$(\mathcal{L} \otimes \mathcal{M})(U) := \mathcal{L}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{M}(U).$$

Moreover, for any invertible sheaf $\mathcal{L}$, there exists an invertible sheaf $\mathcal{L}^{-1}$ such that $\mathcal{L} \otimes \mathcal{L}^{-1} \simeq \mathcal{O}_X$. Namely, $\mathcal{L}^{-1} = \mathcal{L}^\vee := \text{Hom}(\mathcal{L}, \mathcal{O}_X)$ is the dual sheaf.

Hence, the Picard group is the group of invertible sheaves under $\otimes$.

Huybrechts 2.2.1. (Holomorphic vector bundles)

Let X be a complex manifold. A holomorphic vector bundle of rank r on X is a complex manifold E together with a holomorphic map $\pi : E \rightarrow X$ such that:

1. $\pi^{-1}(x)$ is an r -dimensional complex vector space for all $x \in X$;
2. There exists an open cover $\{U_\alpha\}$ of X and biholomorphic maps $\psi_\alpha : \pi^{-1}(U_\alpha) \simeq U_\alpha \times \mathbb{C}^r$, called *trivializations*, that commutes with the transition maps in the sense that the induced maps $\psi_\beta \circ \psi_\alpha^{-1} \in \text{Aut}(U_\alpha \cap U_\beta \times \mathbb{C}^r)$ are $\mathbb{C}$ -linear.

Turns out, a holomorphic vector bundle is exactly specified up to isomorphism by a choice of open cover and transition maps $\psi_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow \text{GL}_r$, whose entries consists of functions on $U_\alpha \cap U_\beta$, that satisfy the *cocycle condition*: $\psi_{\alpha\beta} \circ \psi_{\beta\gamma} = \psi_{\alpha\gamma}$.

A holomorphic line bundle is a holomorphic vector bundle of rank one.

We show that there is a very natural geometric interpretation of locally free sheaves of modules - namely, that they correspond naturally to holomorphic vector bundles.

In particular, invertible sheaves correspond to holomorphic line bundles.

Huybrechts 2.2.18-19 (Locally free sheaf / vector bundle correspondence)

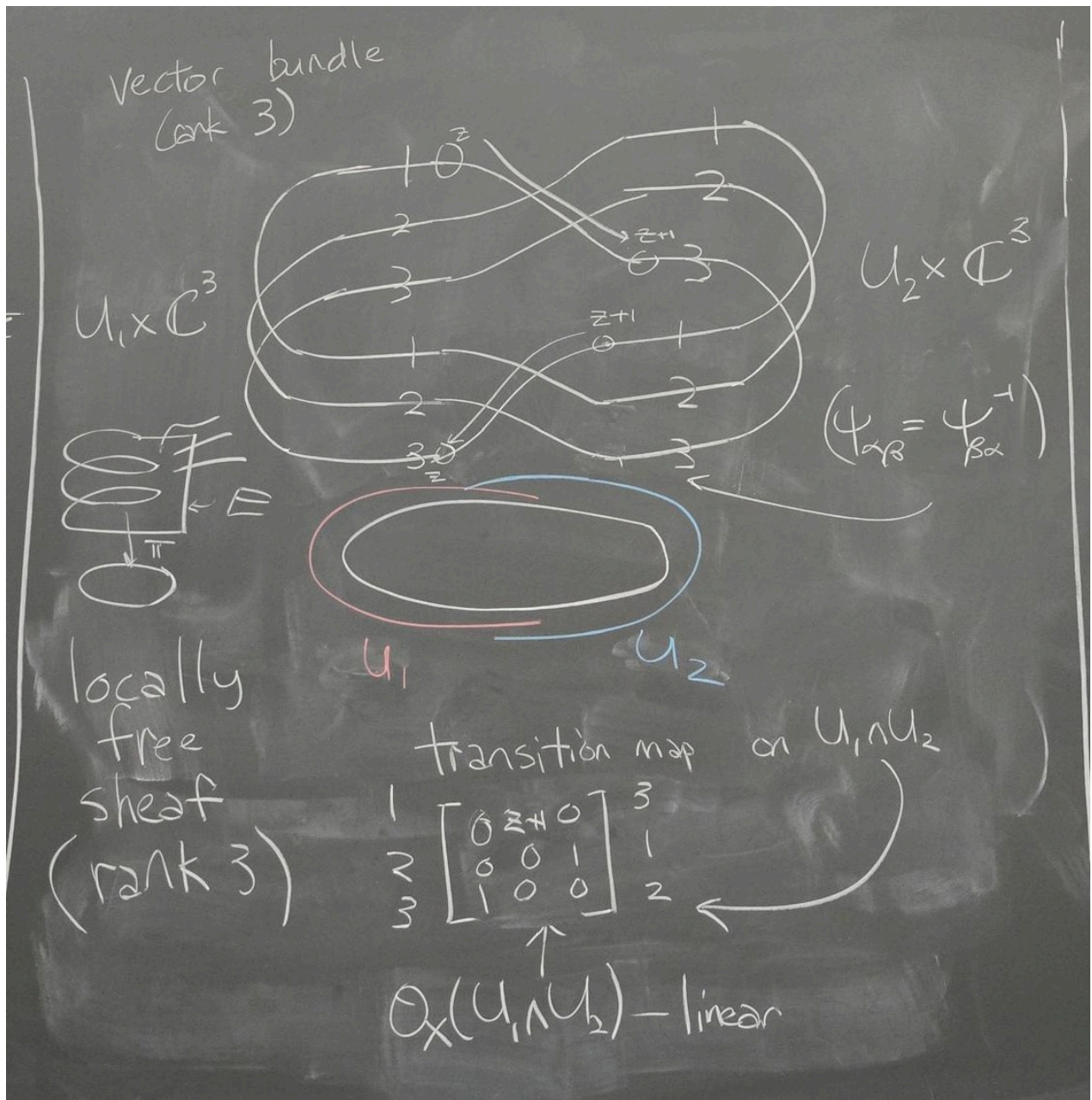
Let $\pi : E \rightarrow X$ be a holomorphic vector bundle.

Then the sheaf of sections, denoted $\mathcal{S}(E/X)$, is the sheaf that associates to U the set of holomorphic sections of π over U . More concretely:

$$\Gamma(U, \mathcal{S}(E/X)) = \{s : U \rightarrow \pi^{-1}(U) \mid s \text{ holomorphic}, \pi \circ s = \text{id}_U\}.$$

This sheaf inherits the structure of a locally free $\mathcal{O}_X$ -module, in the following manner:

- Fix a small enough open set U such that the fiber $\pi^{-1}(U) \simeq U \times \mathbb{C}^r$ is a vector field. Then addition of sections can be defined as addition in the vector field.
- We get natural scalar multiplication as $f \cdot s = f(x) \cdot s(x) \in \pi^{-1}(U)$ for $x \in \mathcal{O}_X(U)$; as $f(x) \in \mathbb{C}$ is a scalar.



On the other hand, let $\mathcal{F}$ be a locally free $\mathcal{O}_X$ -module of rank r .

Suppose that we have chosen trivializations $\psi_\alpha : \mathcal{F}|_{U_\alpha} \simeq \mathcal{O}_{U_\alpha}^{\oplus r}$.

Then the transition maps $\psi_{\alpha\beta}$ are automorphisms of $\mathcal{O}_{U_\alpha \cap U_\beta}^{\oplus r}$ that are $\mathcal{O}_{U_\alpha \cap U_\beta}$ -linear; i.e. elements of $\mathrm{GL}_r(\mathcal{O}_{U_\alpha \cap U_\beta})$ that satisfy the cocycle condition.

- In other words, once you fix a trivializing cover, then the correspondence is directly given by the transition maps: Instead of interpreting the transition maps as isomorphisms between the locally free sections you treat them as the topological gluing condition to get your bundle.
- We give an example of this gluing in action in
- This association gives us a natural bijection between locally free sheaves of rank r and vector bundles of rank r , so we may speak of the two as being the same.
 - Under this interpretation, it is much easier to see why every invertible sheaf has an inverse: you invert the transition maps, and the induced line bundle corresponds to the inverse sheaf!

- This also lends intuition for what multiplication of two vector bundles in the Picard group looks like geometrically: it is just composition of the transition maps. Following this intuition the next three theorems immediately become clear.

Huybrechts 2.3.10,13,14.

1. If D is a (Cartier) divisor given by (f_α, U_α) , then $\mathcal{O}(D)$ is defined to be the line bundle over X with transition functions given by the units $\{f_\alpha f_\beta^{-1}\}$. The transition functions automatically satisfy the cocycle condition so this is indeed well-defined.

This gives a natural group homomorphism $\text{Div}(X) \rightarrow \text{Pic}(X)$ given by $D \mapsto \mathcal{O}(D)$.

- It turns out that this map is exactly the boundary map $H^0(X, \mathcal{K}_X^* / \mathcal{O}_X^*) \rightarrow H^1(X, \mathcal{O}_X^*)$.
2. Let $f : X \rightarrow Y$ be a morphism.
If $D \in \text{Div}(Y)$ is a divisor such that $f^*(D)$ is defined then $\mathcal{O}(f^*(D)) \simeq f^*(\mathcal{O}(D))$.
 3. A divisor $D \in \text{Div}(X)$ is principal if and only if $\mathcal{O}(D) \simeq \mathcal{O}_X$ (the trivial line bundle).
 - This tells us that we in fact have the exact sequence

$$K(X)^* = H^0(X, \mathcal{K}_X^*) \rightarrow H^0(X, \mathcal{K}_X^* / \mathcal{O}_X^*) \rightarrow H^1(X, \mathcal{O}_X^*).$$

- This also tells us that the map $\text{Div}(X) \rightarrow \text{Pic}(X)$ factors through the class group $\text{Cl}(X)$.

Only the proof of the converse of (3) is nontrivial so we include it here.

Proof.

Let $\mathcal{O}(D)$ be trivial. Then Let U_α be an open cover such that D corresponds to (f_α, U_α) . Then there exists holomorphic units $g_\alpha \in \mathcal{O}_X^*(U_\alpha)$ such that $f_\alpha \cdot f_\beta^{-1} = \psi_{\alpha\beta} = g_\alpha \cdot g_\beta^{-1}$. Hence $f_\alpha \cdot g_\alpha^{-1} = f_\beta \cdot g_\beta^{-1}$ on $U_\alpha \cap U_\beta$, so there exists a global meromorphic function f that gives D .

■

We give an example of a nontrivial invertible sheaf.

Vakil 14.1. (Möbius bundle)

Consider the space $\mathbb{P}^1$ (over any field, but we will be taking this as a Riemann surface over $\mathbb{C}$).

Recall that dimension n projective space admits a cover of $n + 1$ copies of dimension n affine space in the natural way.

We define an invertible sheaf on $\mathbb{P}^1$ to be that which is trivial on the open affines

$U_x := \{(x : y) | x \neq 0\}$ and $U_y := \{(x : y) | y \neq 0\}$; with transition function from U_x to U_y given by multiplication by $\frac{x}{y}$.

We compute the global sections of this invertible sheaf:

- A global section is given by a pair $(f(\frac{y}{x}), U_x)$ and $(g(\frac{x}{y}), U_y)$ such that $f(\frac{y}{x}) \cdot \frac{x}{y} = g(\frac{x}{y})$ on $U_x \cap U_y$ and f, g are entire functions.

For simplicity of notation, write $t := \frac{x}{y}$ and $s := \frac{1}{t} = \frac{y}{x}$.

Then $f(t) = \sum_{n=0}^{\infty} a_n t^n$ admits a power series expansion, such that $g(s) = s \cdot f(\frac{1}{s})$.

Then the power series expansion of g is $g(s) = s \cdot \sum_{n=0}^{\infty} a_n s^{-n}$, so we see that only a_0, a_1 can be nonzero.

So the sheaf of global sections are described exactly by the form $(a_0 + a_1 \frac{y}{x}, U_x)$ and $(a_0 \frac{x}{y}, a_1)$ (one can verify easily that every section of this form is well-defined).

Moreover, this implies that this is a nontrivial line bundle:

This invertible sheaf has complex dimension 2, but we know that the sheaf of holomorphic functions on $\mathbb{P}^1$ can only be constant; i.e. have dimension 1.

This sheaf is known as Serre's twisting sheaf, and is denoted $\mathcal{O}(1)$.

- Recall that if $D = \{(f_\alpha, U_\alpha)\}$ and $D' = \{g_\alpha, U_\alpha\}$ then $D + D' = \{f_\alpha g_\alpha, U_\alpha\}$.
As a corollary of [Huybrechts 2.3.10,13,14](#), this implies that $\mathcal{O}(D + D') = \mathcal{O}(D) \otimes \mathcal{O}(D')$.
- This suggestively tells us to define $\mathcal{O}(n)$ over $\mathbb{P}^1$ as n -fold tensor products of $\mathcal{O}(1)$.

Motivated by the result of [Huybrechts 2.2.18-19 \(Locally free sheaf / vector bundle correspondence\)](#), let us try and understand what line bundle this sheaf corresponds to.

For simplicity, we now consider $\mathbb{RP}^1 \simeq S^1$ instead of $\mathbb{P}^1$.

- Then we have a line bundle, trivialized on two halves of the circle.
- When we travel from one half to another, we have to multiply by $\frac{x}{y}$, which flips the sign from $+1$ to -1 .
 - In other words, we do two half-twists when we travel around the unit circle one.
 - This is exactly the Möbius bundle on the unit circle.
- There is a danger: Although it is true that $\mathcal{O}(m)$ corresponds to the m -twist Möbius bundles on the unit circle, it turns out that they are all topologically homeomorphic to either the trivial bundle or the single twist bundle on the unit circle.

Over $\mathbb{C}^n$, we can define the sheaf of degree d polynomials.

However, over $\mathbb{P}^n$, the naïve approach no longer works as $\mathcal{O}(\mathbb{P}^n) = \mathbb{C}$.

We see that the twisting sheaves $\mathcal{O}(m)$ are the natural extension of this over projective space, the line bundle whose sections are homogeneous degree- d "polynomials".

Izadi 1,2. (Twisting sheaves)

Fix an n , and consider $\mathbb{P}^n$.

As before, let U_i denote the affine open set on which x_i is invertible; i.e.

$\{(X_0 : \dots : X_n) \in \mathbb{P}^n \mid X_i \neq 0\}$. This is isomorphic to $\mathbb{C}^n$ as a variety / complex manifold.

We define an invertible sheaf of $\mathbb{P}^n$ as follows:

- On each U_i , consider the free sheaf $\mathcal{O}_{U_i} X_i^m$, i.e. the free rank 1 module with (formal) generator X_i^m . The transition function on the intersection $U_i \cap U_j$ is $(\frac{X_j}{X_i})^m$. With these transition functions, these sheaves glue to the invertible sheaf $\mathcal{O}(m)$ on $\mathbb{P}^n$.

We see that when $m > 0$, global sections of this sheaf are exactly given by degree m homogeneous polynomials, denoted S_m .

Proof.

By a similar argument using power series in [Vakil Ch 14.1. \(Möbius bundle\)](#), elements of this sheaf are always given by polynomials.

- This turns out to be an instance of the [GAGA](#) theorem, which loosely states that the analytic structure sheaf of a compact complex manifold is the same as the algebraic structure sheaf of the manifold as a complex variety.

We define a map $\varphi : \mathcal{O}(m)(\mathbb{P}^n) \rightarrow S_m$ and its inverse $\psi : S_m \rightarrow \mathcal{O}(m)(\mathbb{P}^n)$.

- Suppose that $s \in \mathcal{O}(m)(\mathbb{P}^n)$. For all i , we have the restriction

$$s_i := s|_{U_i} \in \mathcal{O}(m)(U_i) = \mathbb{C} \left[\frac{X_0}{X_i}, \dots, \frac{X_n}{X_i} \right] X_i^m.$$

Let f_i be the polynomial in $R[\frac{X_0}{X_i}, \dots, \frac{X_n}{X_i}]$ such that $s_i = f_i X_i^m$. Since s_i has to glue to s_j on $U_{ij} := U_i \cap U_j$, we have $f_i(\frac{X_0}{X_i}, \dots, \frac{X_n}{X_i}) X_i^m = f_j(\frac{X_0}{X_j}, \dots, \frac{X_n}{X_j}) X_j^m$.

Therefore $P := f_i(\frac{X_0}{X_i}, \dots, \frac{X_n}{X_i}) X_i^m$ is a homogeneous polynomial of degree m , independent of the choice of i and we define $\varphi(s) := P$.

- Conversely, starting with $P \in S_m$, for all i , we can write $P = f_i(\frac{X_0}{X_i}, \dots, \frac{X_n}{X_i}) X_i^m$, which defines a section s_i of $\mathcal{O}_{U_i} X_i^m$. By definition, $s_i|_{U_{ij}} = s_j|_{U_{ij}}$, and the s_i glue to define the global section $\psi(P)$ of $\mathcal{O}(m)$. ■

It turns out that $\mathcal{O}(m)$ gives the degree- n Veronese embedding of $\mathbb{P}^1$ into $\mathbb{P}^m$.

One can show, that $\mathcal{O}(m)^{-1} \cong \mathcal{O}(-m)$, $\mathcal{O}(m) \otimes \mathcal{O}(p) \cong \mathcal{O}(m+p)$ and that the sheaves $\mathcal{O}(m)$ and $\mathcal{O}(p)$ are not isomorphic for $m \neq p$. So the sheaves $\mathcal{O}(n)$ form a free abelian group of rank 1, i.e., $\mathbb{Z}$.

It turns out all invertible sheaves are of this form, such that $\text{Pic}(\mathbb{P}^n) \simeq \mathbb{Z}$.